

Umer Niazi

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EDUCATION

University of Engineering and Technology, Lahore

BS in Artificial Intelligence

Lahore, Pakistan

2025 – 2029 (*Expected*)

Forman Christian College

Intermediate in Computer Science (ICS)

Lahore, Pakistan

2022 – 2024

PROJECTS

Thermal Capital | *Python, FastAPI, React, TypeScript, Geospatial Processing, SQLite*

Aug 2026

- Developed a decision-support prototype for municipal heat-intervention planning that pairs geospatial processing with constrained optimization to maximize urban cooling within capital budget limits
- Engineered a FastAPI REST API and interactive React/TypeScript interface to simulate cooling interventions, evaluate spatial coverage, and optimize multi-asset municipal budgets
- Implemented 112 automated unit and integration tests across data models, knapsack optimization algorithms, and API endpoints to ensure system correctness and reproducibility

News NLP Pipeline | *Python, BERTopic, spaCy, Transformers, Streamlit, SQLite*

May – July 2026

- Built an end-to-end NLP pipeline for unsupervised topic discovery, sentiment analysis, and named-entity extraction across 15 years of Dawn News headlines without manual labeling
- Scraped and preprocessed 350K+ headlines and 420K+ entity mentions using BERTopic and spaCy; modeled 20 distinct topic clusters and deployed an interactive Streamlit dashboard

Himalayan Wildlife Classifier | *Python, PyTorch, ResNet18, Streamlit*

May – July 2026

- Built an image classifier for Snow Leopard, Markhor, and Himalayan Brown Bear, plus a background Other class, using transfer learning on a frozen ResNet18 backbone
- Achieved 96.55% validation accuracy; implemented an 80% confidence threshold for selective prediction to filter ambiguous inputs and deployed an interactive Streamlit demo

wmus | *Python, windows-curses, pygame*

Jan – June 2026

- Built a native Windows terminal music player inspired by the cmus workflow, featuring keyboard-first navigation, a custom curses interface, and configurable keybindings
- Implemented fuzzy track search and metadata caching for fast startup across large audio collections; packaged as a standalone installer using PyInstaller and Inno Setup

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL

AI & Machine Learning: PyTorch, scikit-learn, Transformers, Computer Vision, NLP, Hugging Face, spaCy, BERTopic

Web & Software Engineering: FastAPI, React, Tailwind CSS, REST APIs, Streamlit

Data & Systems: Pandas, NumPy, SQLite, Linux, Git, Bash